OPEN AGENT HACKATHON 2026 · TRACK 02 · THE AGENT THAT CAN EXPLAIN WHY

# x402 Adoption Investigator

A multi-agent system that answers "how real is x402 adoption?" and proves every number it says: each claim links to a URL with the hash of the bytes fetched, a Base transaction, or a line of its own log.

x402 = pay-per-call over HTTP 402 Payment Required, settled in USDC on Base.

Team Automata · Martin Pisacreta (solo, Argentina) · Zetaris · Meterless World Model, H-MEM and Nightshift · NVIDIA Nemotron · built with Claude Code and Cursor Agent

THE USER AND THE PROBLEM

# One store, five sources, five answers

User: the operator of an autonomous agent that sells or buys over x402 and must decide where the real demand is. Asked how many paid endpoints one store has, three sources say 17, two say 19 and 36, and none of them says why.

| SOURCE | WHAT IT MEASURES | ENDPOINTS FOR TIENDA.TEXTOKIT.STORE |
|---|---|---|
| CDP Bazaar search | listed resources, capped at 20 results | **17** |
| Store openapi.json | declared paid operations | **17** |
| Agent402 | paid tools (of 34 tools) | **17** |
| vet402 | resolved endpoints | **19** |
| x402-trust | keyword search matches | **36** |

Activity disagrees too: Bazaar 30-day counters sum to 60 calls, the seller's receipts list 77 settled sales, and 56 of them were paid by the seller itself.

WHY THIS NEEDS AGENTS

# A script would pick one number. This system explains all of them.

### Live, conflicting sources

Bazaar, the seller's receipts, Base RPC, x402-trust, Agent402 and vet402 each measure a different thing on a different window.

### On-chain verification

Settlements are checked on Base by decoding the USDC Transfer log, never trusting tx.from, which is the facilitator.

### Disagreement as output

When sources differ, the answer carries a Conflict object with each value, its measurement basis and why they differ.

### Failure is normal

Sources go down mid-run. They become explicit "unavailable" entries with a logged recovery, never silent zeros.

WHAT GOES IN, WHAT COMES OUT

# A question in, a proof-carrying answer out

Input: one JSON question

```
{"run_id": "ex01-seller-audit",
 "input": {"question": "How real is the adoption of
   tienda.textokit.store? Who pays, how much,
   do they come back, and why do registries
   disagree about it?"}}
```

Output: one JSON envelope and a Markdown report

- **verdict** chosen by rules, never by a model
- **claims[]** each with evidence: URL + sha256, Base tx, or log line
- **conflicts[]** with per-source values and the explanation
- **rejected_claims[]** with the critique trail
- **unavailable[]** instead of zeros

ARCHITECTURE

# Seven roles, bounded loops, every handoff logged

**Planner**
picks the playbook, sources and queries

**Collector**
fetches every source and hashes the bytes

**Keeper**
folds evidence into the World Model

**Verifier**
decodes settlements on Base

**Analyst**
computes each metric in SQL and in Python

**Writer**
drafts an answer that cites claims

**Critic**
rules C1 to C7: rejects or sends back

## Computed twice

Every metric has a committed SQL query and a pure Python reducer over the same exports. A number survives only if both exist and agree (rule C2); a mismatch stays as an open Conflict.

## Bounded and logged

Deadline, fetch, SQL, model-call and critic-round budgets. Every handoff, fetch, sql, generate, claim, conflict, critique, retry and recovery line goes to logs/<run_id>/run.jsonl.

> Planner, Writer and the Critic's second opinion send their prompts to Meterless Nightshift. With the model pinned (METERLESS_MODEL), NVIDIA Nemotron answered them in the live runs of 24 Oct, and sample mode replays 25 Nemotron answers recorded through Nightshift on the fixture prompts; with the default meterless/auto the route may differ, and every receipt names the model. The model writes the plan's untested hypotheses, the drafts and extra critique issues; it never accepts a claim.

A REAL RUN, CONDENSED

# What a judge sees in run.jsonl

```
guard     network disabled in sample mode (socket.connect patched)
generate  planner    nvidia/nemotron-3.5-lightning-30b-a3b (recorded answer)
handoff   planner → collector   plan (10 items, sha256 abce43a7…)
recovery  collector  fixture_miss on a Base receipt: settlement marked unverifiable
handoff   keeper → analyst      query_ids (13)
conflict  k01 endpoint_count_for_host   five sources, five measurement bases
conflict  k02 calls_30d_for_host        Bazaar 30-day counter vs seller receipts
generate  writer     writer-r1 draft (recorded answer)
critique  round 1  C1…C7 accept 19 claims · C3 revise ×19: no caveat codes · B revise '15'
critique  round 1  LLM 10 issues (reply cut at 500 tokens) · 5 dismissed: rule B backs them
critique  round 2  C3 revise ×19 · C7 revise: k02 not named · B revise '15' · LLM 5
recovery  writer     1 flagged sentence cut · conflicts, caveats, verdict added by code
result    ok · 19 accepted · 0 rejected · 2 conflicts · 2 critic rounds
```

Condensed from the 182 JSON lines of example 01: 13 handoffs, 17 fetches, 13 SQL queries, 5 Nemotron requests replayed from recorded answers, 70 critiques, 9 recoveries, 2 conflicts.

CRITIQUE AND RECOVERY

# The Critic says no, and the answer says why

```
claim      c07 verified_tx = 0   sql 0 · reducer 0
critique   round 1  C1 reject c07: no decoded
           settlement of 0xa1d1…1978 (1 sampled:
           unverifiable, fixture_miss)
critique   round 1  C3 revise ×6 · B revise: sentence 5
           cites c07, sentences 5 and 7 carry its 0
critique   round 1  LLM 9 issues · 4 dismissed by B
handoff    critic → writer   issues (16)
recovery   writer  narrative_trimmed: c07 left
           out of the answer and named
critique   round 2  C1…C7 accept 6 · C3 revise ×6 · B accept
recovery   writer  3 sentences cut · caveats and
           verdict sentence added by code
result     partial · 6 accepted · 1 rejected
output     verdict insufficient_evidence (by rules)
```

### Why the zero is rejected

SQL and reducer agree on 0, but nothing was decoded for this wallet. Zero verified would be a number about missing data, so the Critic reports it as unknown.

### Not staged

Same question as example 03, another external payer of the same store. Live, its settlement is verified at block 52,914,075.

Condensed from the 121 JSON lines of example 04: 13 handoffs, 49 critiques (17 from Nemotron's second opinion, 9 of them dismissed by rule B), 10 recoveries, 2 critic rounds.

METERLESS, STRUCTURAL

# World Model as shared state, H-MEM as policy

## World Model

- Entities: sellers, endpoints, wallets, facilitators, registries, with bitemporal relationships
- Provenance on every row; a claim without a URL, tx or log line is refused by the engine
- Idempotent ingest, proved in every output: the second pass appends 0 events
- 8 of 8 conformance checks

## H-MEM

- Seven pitfall memories ("never sum unique payers across endpoints", "receipts are self-reported") add caveats that Critic rule C3 checks
- Injected into Planner and Writer prompts when a model is mounted, logged by hash; tested offline
- Ablation committed: one memory less drops a caveat and fifteen citations, not the verdict
- 13 of 13 upstream vectors

METERLESS NIGHTSHIFT, THE AI PROVIDER · NVIDIA NEMOTRON

# Pinned to Nemotron, every request's receipt names it

| REQUEST (LIVE, EXAMPLE 01, 24 OCT 18:55 UTC) | ROUTED_MODEL IN THE COMPUTE RECEIPT | KIND | RECEIPT | MS |
|---|---|---|---|---|
| planner | `nvidia/nemotron-3.5-lightning-30b-a3b` | cloud | `rcpt_req_t1f1teb8` | **1,955** |
| writer-r1 | `nvidia/nemotron-3.5-lightning-30b-a3b` | cloud | `rcpt_req_c0jovllr` | **5,990** |
| critic-r1 | `nvidia/nemotron-3.5-lightning-30b-a3b` | cloud | `rcpt_req_pl9z663t` | **4,236** |
| writer-r2 | `nvidia/nemotron-3.5-lightning-30b-a3b` | cloud | `rcpt_req_v4ma9ufl` | **12,867** |
| critic-r2 | `nvidia/nemotron-3.5-lightning-30b-a3b` | cloud | `rcpt_req_y94ho7dr` | **4,714** |

### Hosted, primary

Nemotron 3.5 Lightning 30B A3B on NVIDIA's API, reached by Nightshift with the key set in its Providers tab; the app never sees it. 11 live runs (5 on Zetaris), 57 requests, all kind cloud.

### Local GPU

Nemotron Nano 9B v2 (Q4_K_M), every layer on the RTX 5080 Laptop GPU. 2 live runs (1 on Zetaris), 10 requests, kind local, privacy local-only.

### Offline

Sample mode replays 25 recorded Nemotron 3.5 answers, keyed by the sha256 of each prompt: no keys, no network. Thinking is off per request (reasoning_effort none, /no_think).

Pinned: METERLESS_MODEL=nvidia/nemotron-3.5-lightning-30b-a3b. Nightshift's own signed ledger holds these five receipts, each Ed25519 signature valid. With the portable default meterless/auto the route may differ: the six earlier live runs (32 requests) and the first replay set (25) went to a local Llama 3.2 3B lane, and their receipts say so (archived). Zetaris was stopped for the run above (18:55 UTC), so its SQL ran on SQLite.

ZETARIS, THE LIVE FEDERATION LAYER

# Zetaris when it is up, SQLite otherwise, same rows

- A live run with Zetaris reachable registers its exports as Lightning REST tables (receipts, Bazaar, openapi, Agent402, vet402, x402-trust, settlements) and flattens them with SchemaStore views (the 22 Oct live traces, with no model, and six runs of 24 Oct with Nemotron); without it the same SQL runs on SQLite
- The live seller audit computed all 19 claims on Zetaris 2.4.3.1
- Offline, the same SQL runs on SQLite; a test proves equal rows against the 19 result sets Zetaris recorded

### Logged per query

Executor, SQL sha256, rows and time in sql.jsonl, with the run's tables and their source URLs; every REST call in zetaris.jsonl. --require-zetaris turns any fallback a query uses into an error.

### Zetaris and Nemotron in one run

Six live runs of 24 Oct, 22:36 to 22:42 UTC, ran every query on Zetaris (q20 to q22 and chain_facts included) while Nemotron answered through Nightshift: status ok. Earlier that day Zetaris was stopped and the Nightshift runs used SQLite.

RESULTS ON REAL DATA

# Four questions, four honest answers

### Seller audit · self_settled_mostly

77 sales, 56 paid by the seller's own wallets; 21 external sales by 4 payers, 0.044 USDC in total.

### Catalog · concentrated

28,462 Bazaar resources on 2,680 hosts; the top host holds 29.5% of 824,796 calls in 30 days.

### Payer dossier · adoption_thin

One purchase of 0.001 USDC, its settlement decoded and verified on Base against the receipts.

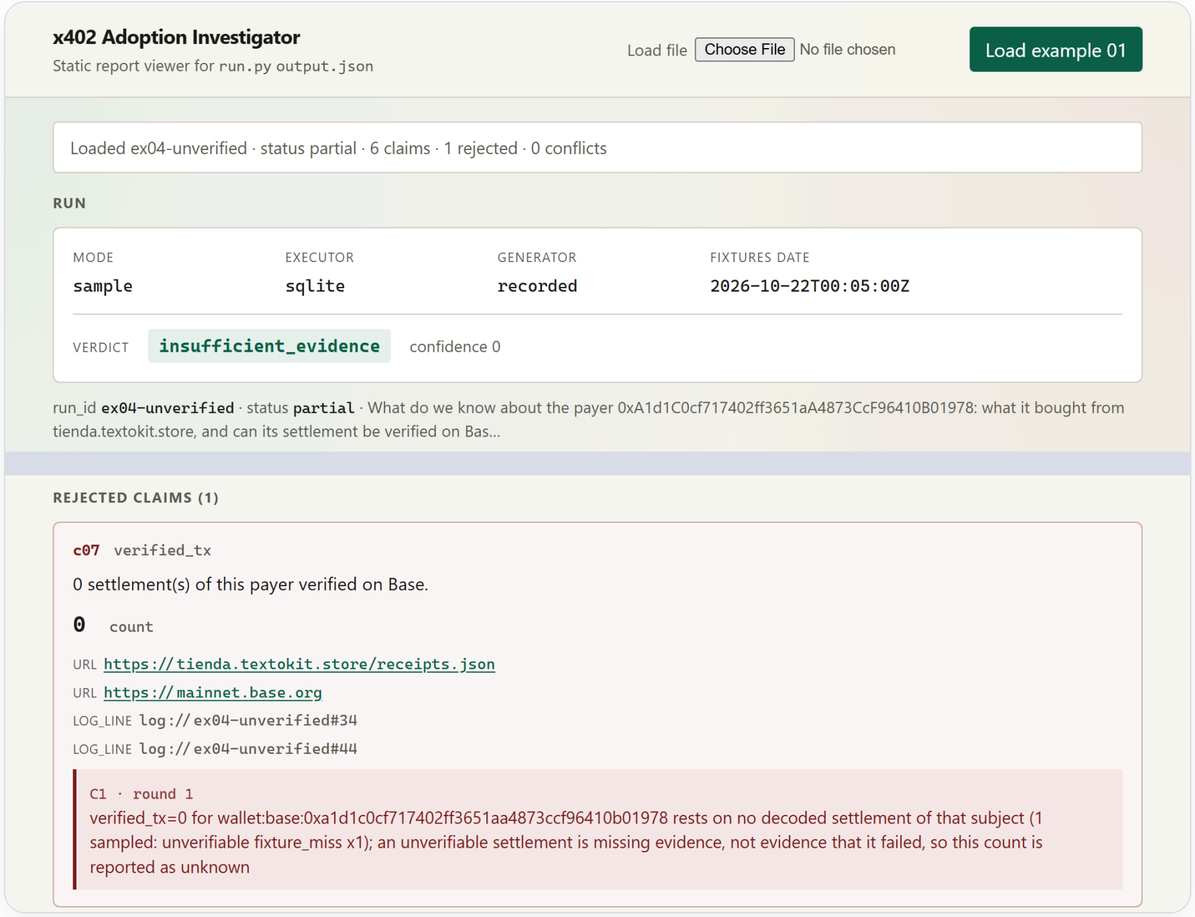
### Unverifiable payer · insufficient_evidence

One purchase of 0.001 USDC; "0 verified" rejected by the Critic, balance and nonce unavailable.

The honest finding for our own store: most of its sales were paid by its own wallets. The system says so, with the proof.

SEE IT

# The report viewer




## Any output.json

Run header with the mounts, accepted claims with their evidence links, rejected claims with the Critic's reason, and conflicts with each source's value and explanation.

Static HTML, no build step, no CDN. Shown: example 04 in sample mode, two crops of one page (header, rejected claims).

STACK AND HOW TO TRY IT

# One command, offline, no keys

```
git clone https://github.com/martinPisacreta/x402-adoption-investigator
cd x402-adoption-investigator
python run.py --sample --input examples/ex4.json
```

**Python 3.11, standard library** **Zetaris Lightning 2.4.3.1** **Meterless World Model + H-MEM**

**Meterless Nightshift 1.2.3: primary AI provider** **NVIDIA Nemotron 3.5 Lightning, hosted** **NVIDIA Nemotron Nano 9B v2, RTX 5080**

**Base JSON-RPC** **Docker** **330 tests** **Claude Code: main coding agent** **Cursor Agent: viewer, README pitch draft**

The Nemotron live runs of 24 Oct, model pinned, sent 57 requests to Nemotron 3.5 Lightning and 10 to Nemotron Nano 9B v2 through Nightshift (13 runs, six of them on Zetaris), one compute receipt each. The command above replays recorded Nemotron answers, offline and without keys.

Pre-existing, declared: our own Automata agent and its x402 store tienda.textokit.store (data source only, none of its code here), the Meterless conformance vectors (copied with attribution), the Zetaris platform, Meterless Nightshift 1.2.3 with its llama.cpp server and the models it served (Nemotron 3.5 Lightning hosted by NVIDIA, Nemotron Nano 9B v2 as bartowski's Q4_K_M GGUF, Llama 3.2 3B Q8; no code or weights of theirs here) and the Python standard library.

WHAT WAS UNIQUE, AND WHAT COMES NEXT

# The model writes, the rules decide, the logs prove it

## What was unique

### Rules decide, the model drafts

Nemotron writes the plan's hypotheses (untested), the answer and a second-opinion critique, whose issues can only get sentences revised or cut. Claim values, rejections and the verdict come from SQL, reducers and rules (C1 to C7, the verdict rules), never from the model.

### The model's critique is checked

An issue on a sentence whose numbers rule B maps to the claims it cites is dismissed and logged: 20 of the 58 model issues in the five examples.

### Every number twice, a receipt for every answer

SQL and a Python reducer over the hashed evidence must agree (C2). Sample mode replays real Nemotron answers by prompt sha256. All 157 cited receipts are in the committed excerpt of Nightshift's ledger, signatures valid: unchanged since signing, not which machine signed.

## Future enhancements

### Why the Zetaris heap ran out

Zetaris and Nightshift now run together (six live runs of 24 Oct, status ok), but the 2 GB driver heap ran out once in those runs, caching a Bazaar page, for a reason not isolated. Next: a retry and a larger heap, and recorded Zetaris result sets for q20 to q22.

### A full live run from the container

Today one Nightshift request from Docker is committed; next, the investigator-live compose service end to end.

### Wider data, better drafts

More stores' receipts and settlements sampled across the Bazaar; prompts that make Nemotron keep caveat codes and conflict ids, which the code adds today.